

CAPSTONE PROJECT

Insurance Data Analysis Project

Project Overview

This project demonstrates how data science is used in healthcare to analyze medical insurance data and understand the factors that influence healthcare costs. The analysis focuses on identifying relationships between demographic characteristics, lifestyle habits, and insurance charges to generate meaningful insights for healthcare decision-making and predictive modeling.

The project uses a structured healthcare insurance dataset containing demographic information, health indicators, family coverage details, geographic location, and medical insurance expenses. Through data cleaning, exploratory data analysis (EDA) and visualization, the project aims to uncover patterns that explain variations in healthcare costs among insured individuals.

Project Objectives

The project seeks to:

- Clean and preprocess healthcare insurance data for analysis.
- Explore patterns and trends in medical insurance charges.
- Investigate how demographic factors such as age and gender influence healthcare expenses.
- Examine the impact of lifestyle factors such as smoking habits and Body Mass Index (BMI) on insurance costs.
- Analyze family-related variables such as the number of dependents covered by insurance.
- Compare insurance charges across different geographic regions.

Dataset Description

The Healthcare Insurance Dataset contains structured tabular data describing individual characteristics and lifestyle factors that may affect insurance costs.

The dataset includes the following variables:

Variable	Description
Age	Age of the insured person
Sex	Gender of the insured individual
BMI	Body Mass Index, indicating body fat based on height and weight
Children	Number of dependents covered under the insurance policy
Smoker	Indicates whether the individual is a smoker
Region	Geographic region of insurance coverage
Charges	Total medical insurance costs incurred

Data Science Workflow

The project follows a standard healthcare analytics workflow:

1. **Data Collection and Loading**
Import the healthcare insurance dataset into Python for analysis.
2. **Data Cleaning and Preprocessing**
Handle missing values, remove duplicates, encode categorical variables, and prepare the data for modeling.
3. **Exploratory Data Analysis (EDA)**
Explore variable distributions, detect trends, and investigate relationships between predictors and insurance charges.
4. **Data Visualization**
Use charts, histograms, boxplots, and heatmaps to visualize healthcare patterns and cost drivers.
5. **Interpretation and Insights**
Identify major healthcare cost determinants and interpret findings for healthcare analytics and decision-making.

Expected Outcomes

By completing this project, users will be able to:

- Understand how healthcare insurance data is analyzed using data science methods.
- Perform professional exploratory data analysis in Python.
- Build predictive models for healthcare cost estimation.
- Interpret relationships between risk factors and insurance expenses.
- Generate actionable insights for cost prediction and risk assessment.

Applications of the Project

This project has practical applications in:

- **Healthcare analytics**
- **Medical cost prediction**
- **Insurance risk assessment**
- **Healthcare policy analysis**
- **Predictive modeling**
- **Health economics research**
- **Data science and machine learning education**

Dataset: [Insurance dataset](#)